

7.1 Analytical Question: Asymptotic Properties of MLE

Problem (unchanged).** Define and explain the significance of: Consistency, Asymptotic Normality, Efficiency.**

Consistency

Definition. An estimator $\hat{\theta}_N$ is **consistent** for the true parameter θ_0 if

$$\hat{\theta}_N \xrightarrow{p} \theta_0 \quad (N \rightarrow \infty),$$

i.e. for every $\varepsilon > 0$, $\Pr(|\hat{\theta}_N - \theta_0| > \varepsilon) \rightarrow 0$.

Significance. Consistency guarantees that with enough data the estimator converges to the true parameter. For ML practitioners, it formalizes the intuition that “more data $\rightarrow$ better parameter estimates” (under model correctness and standard regularity conditions).

Asymptotic Normality

Formal statement. Under regularity conditions,

$$\sqrt{N} (\hat{\theta}_N - \theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1}),$$

where $I(\theta_0)$ is the Fisher information per sample (so the total information is $NI(\theta_0)$).

Significance. Asymptotic normality gives a practical approximation to the sampling distribution of $\hat{\theta}_N$ for large N . It enables:

- **Confidence intervals:** $\hat{\theta}_N \pm z_{\alpha/2} \sqrt{I(\theta_0)^{-1}/N}$.
- **Wald tests** and approximate hypothesis tests.
- Quantifying uncertainty and comparing models or parameters.

Efficiency (Asymptotic Efficiency)

Definition. The Cramér–Rao lower bound (CRLB) says that any unbiased estimator $\tilde{\theta}$ must satisfy

$$\text{Var}(\tilde{\theta}) \geq \frac{1}{NI(\theta_0)}.$$

An estimator is **asymptotically efficient** if its asymptotic variance equals this lower bound, i.e. the MLE achieves

$$\text{asympt. Var}(\hat{\theta}_N) = \frac{1}{NI(\theta_0)}.$$

Significance. Efficiency means the estimator makes the “best possible” use of data asymptotically — no (regular) unbiased estimator has lower asymptotic variance. In practice, MLEs often attain this bound under standard regularity conditions, so they are asymptotically optimal.

A short note on regularity conditions

All three properties require standard regularity assumptions (identifiability, smoothness of the likelihood, dominated convergence allowing interchange of differentiation and expectation, interior true parameter, finite Fisher information, etc.). If those fail (e.g., model misspecification, boundary parameters, heavy tails), some properties may not hold.

7.2 Numerical Problem: MLE for the Laplace Location (with known scale b)

Model. For known $b > 0$,

$$f(x \mid \mu, b) = \frac{1}{2b} \exp(-|x - \mu|/b).$$

Given i.i.d. data $D = \{x_1, \dots, x_N\}$, find the MLE $\hat{\mu}$.

1) Likelihood and log-likelihood

$$L(\mu) = \prod_{i=1}^N \frac{1}{2b} \exp(-|x_i - \mu|/b) = \left(\frac{1}{2b}\right)^N \exp\left(-\frac{1}{b} \sum_{i=1}^N |x_i - \mu|\right).$$

Log-likelihood (up to constants):

$$\ell(\mu) = -\frac{1}{b} \sum_{i=1}^N |x_i - \mu| + \text{const.}$$

2) MLE = minimizer of sum of absolute deviations

Maximizing $\ell(\mu)$ is equivalent to minimizing

$$g(\mu) = \sum_{i=1}^N |x_i - \mu|.$$

It is a standard fact that any median of the sample minimizes $g(\mu)$. Thus

$$\hat{\mu}_{\text{MLE}} = \text{sample median of } \{x_i\}$$

(with the usual convention for even N : any point in the interval between the two center order statistics minimizes the objective; the midpoint is commonly taken).

Sketch via subgradient/first-order condition. The subgradient of $|x_i - \mu|$ wrt μ is $-\text{sgn}(x_i - \mu)$ (with multivalued sgn at zero). The subgradient condition for a minimum is

$$0 \in - \sum_{i=1}^N \text{sgn}(x_i - \mu),$$

i.e. the number of observations above μ equals the number below μ (definition of median).

3) Asymptotic variance and efficiency (nice closed form)

We can compute the Fisher information per sample for μ (with known b).

Score for one observation:

$$s(x; \mu) = \frac{\partial}{\partial \mu} \log f(x | \mu) = \frac{1}{b} \text{sgn}(x - \mu).$$

(At $x = \mu$ the derivative is undefined, but this set has probability zero under continuous densities.)

Fisher information per sample:

$$I(\mu) = E[s(x; \mu)^2] = E\left[\frac{1}{b^2} \text{sgn}(x - \mu)^2\right] = \frac{1}{b^2},$$

since $\text{sgn}^2(\cdot) = 1$ almost surely. For N i.i.d. samples, total information is $NI(\mu) = N/b^2$.

CRLB gives minimum asymptotic variance $1/(NI(\mu)) = b^2/N$.

The sample median (the MLE here) has asymptotic variance equal to the CRLB:

$$\sqrt{N} (\hat{\mu} - \mu) \xrightarrow{d} N(0, b^2), \quad \text{so} \quad \text{Var}(\hat{\mu}) \approx \frac{b^2}{N}.$$

Hence the Laplace-location MLE is **asymptotically efficient**.

(Brief reasoning: for a density with $f(\mu) > 0$, the asymptotic variance of the sample median is $1/(4Nf(\mu)^2)$. For Laplace, $f(\mu) = 1/(2b)$, so $1/(4Nf(\mu)^2) = 1/(4N \cdot 1/(4b^2)) = b^2/N$.)

Quick summary (one-liners)

- **Consistency:** MLE $\rightarrow$ true parameter as $N \rightarrow \infty$.
- **Asymptotic normality:** $\sqrt{N}(\hat{\theta} - \theta_0) \rightarrow N(0, I(\theta_0)^{-1})$.
- **Efficiency:** MLE attains CRLB asymptotically; no regular unbiased estimator has lower asymptotic variance.